

SUJITH MAKAM

AI Developer | Software Engineer

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Leet Code

Professional Summary

AI and Cloud developer who builds and ships LLM, agentic, and generative-AI systems into production on Azure and AWS. As a Software Engineer at CGI, I develop enterprise GenAI and multi-agent solutions across Azure AI Foundry, Azure OpenAI, and AWS, backed by RAG pipelines, evals, and Infrastructure-as-Code, delivering measurable outcomes (85% fewer agent escalations, 40,000+ calls auto-evaluated); 4 IEEE publications in applied ML.

Education

Indian Institute of Information Technology (IIIT), Dharwad

B.Tech in Computer Science and Engineering

JULY 2025

CGPA: 8.41

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, C, SQL

AI & Machine Learning: Generative AI, Agentic AI, LLMs, RAG, Multi-Agent Systems, LLM Evaluation, Prompt Engineering, MCP, A2A Protocol

AI Frameworks & Platforms: Amazon Bedrock, Vertex AI, Azure AI Foundry, Google ADK, CrewAI, AWS Strands, Cognigy, Genesys

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Pydantic, Microservices, Serverless, API Design

Cloud & DevOps: AWS (Lambda, CDK, S3, EventBridge, IAM, KMS, StepFunctions, SNS), Azure (Functions), GCP (Cloud Run, Dialogflow CX), Docker, CI/CD, Infrastructure as Code, Distributed Systems

Databases: PostgreSQL, DynamoDB, Cosmos DB, MongoDB, Data Modelling

Observability: OpenTelemetry, Langfuse, AgentOps, Distributed Tracing, CloudWatch

Frontend & Tools: React.js, Next.js, Redux, Tailwind CSS, Git, Linux

Experience

CGI

Jul. 2025 - Present

Software Engineer

- Designed and deployed enterprise **Conversational AI** and **Agentic AI** solutions, building task-specific agents with **Azure AI Foundry, Google ADK, and CrewAI** and integrating Azure OpenAI, Cognigy, the **Genesys** CCaaS platform, and AWS to automate customer-support workflows, **reducing Average Handle Time (AHT) by 3–7 minutes**.
- Built an **agentic AI QA platform** on **Amazon Bedrock** that scores **100% of call transcripts against 24 quality parameters, automating 90%+ of QA** that was previously manual and sampled only a fraction of calls; later ported the pipeline to **Azure Functions and Cosmos DB**.
- Engineered end-to-end **Infrastructure-as-Code** (AWS CDK/TypeScript, AWS SDK) to provision multi-region **Amazon Connect** environments (**80% less manual setup**) and a production **disaster-recovery sync** (boto3, Lambda, EventBridge) holding a cross-region mirror at **100% config compliance** with zero-downtime failover, delivered with **AI-accelerated development**.

Indian Institute of Science Bangalore

May. 2024 - July. 2024

Research Intern

- Investigated LLM Pruning techniques for improving model efficiency, optimizing the attention and feed-forward layers of GPT-2. Achieved a 5.2% reduction in instruction cache misses through tiled matrix multiplication implementation.
- Conducted detailed profiling of GPT-2 using Intel VTune, identifying bottlenecks and enhancing cache efficiency, which resulted in a 4x speedup in attention layer execution.
- Optimized identified bottlenecks in the attention and feed-forward layers of GPT-2 and Reduced memory bandwidth usage by 34.8 %, decreasing memory-bound issues and ensuring more efficient execution
- Implemented custom classes (e.g., tiled matrix multiplication) for performance improvement in the attention layer.

Projects

AI Governance Platform | ADK, Foundry, MCP, A2A, Langfuse



- Built an enterprise **AI Governance Platform** for secure orchestration of **multi-agent LLM** workflows using **Azure AI Foundry, Google ADK, CrewAI**, and Agent Gateway.

- Added end-to-end observability with **Langfuse and AgentOps** (prompt tracing, KPI and latency monitoring, production evaluation) and **Responsible AI** controls: prompt-injection prevention, sensitive-data masking, virtual API isolation, policy enforcement, and secure tool execution.

LegacyLift | Python, FastAPI, MCP, A2A, Next.js

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- Built a **multi-agent** system that makes legacy **COBOL** logic callable by AI agents without rewriting the mainframe: 5 agents (Manager, Analyzer, Developer, QA, Feasibility) coordinating over an **A2A message bus** with a shared context graph. **Shortlisted top 13 of 200+ entries at CGI Envision 2026.**
- Designed a two-phase rollout: **Phase 1** ships an **MCP wrapper** that AI-enables users from day one, while **Phase 2** modernizes the codebase component-by-component via the **strangler-fig cut-over pattern** — with **human-in-the-loop approval gates** at every step of both phases.
- Parsed **COBOL** source to identify exposable routines and auto-generated **MCP tools** around them; used a provider abstraction to avoid vendor SDK coupling and streamed live execution state to a **Next.js** UI over a **FastAPI WebSocket**.

Certifications

- Full Stack Web Development from Udemy.
 - Machine Learning A-Z from Udemy.
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Research Publications (IEEE)

- **COMSNETS 2026:** FAPL-DM-BC: Privacy-Aware, Blockchain-Backed FL Framework with Dual-Model XAI for IoVs.
- **IEEE FNWF 2025:** Enhancing IoT Network Security through Adaptive Curriculum Learning and XAI.
- **IEEE FNWF 2025:** FedGuard-IIoT: A Scalable Federated Learning Architecture for Network-IDS in the Industrial IoT.
- **IEEE SPACE 2025:** Predicting Stellar Metallicity: A Comparative Analysis of Regression Models for Solar Twin Stars.

Achievements

TCS TechBytes: Dharwad Regional Finals Runner Up
A Prestigious Tech Quiz conducted by TCS and BITES in Karnataka

State Rank Holder: Andhra Pradesh State Board Topper
Scored 990/1000 in my state board examinations